

BORIS LIGHTING

DECORATIVE LED STREET LIGHTING SYSTEMS



STREET LIGHT SUBMITTAL PACKAGE

LP-2A Collector / LP-3A Arterial — DRAFT 1

PROJECT: Municipal Road Lighting — Saratoga Springs, UT
SUBMITTED BY: Boris Lighting
DATE: March 2026
SUBMITTAL NO: SS-2026-LP-001 / DRAFT 1
STANDARD: Saratoga Springs City Engineering Standard 05540

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PRODUCTS SPECIFIED

LUMINAIRE: Boris DSDP25 — Solid State Decorative Pendant, 80 PLED, 120V, 3000K, 70.49W
ARM: Boris XAS-2-180 — Double Pendant Arm, Black RAL-9005-T
POLE: Boris RNTA205-188 — 20' Round Straight Steel Pole, 5" nominal, Q235 steel
BASE: Boris Decorative Cast Aluminum Base, 2-piece wraparound, Black RAL-9005-T
DRIVER: Signify Advance Xitanium XI180C090V285BSF1, 0–10V, UL 8750

This submittal package has been prepared for review by the City of Saratoga Springs. All data traced to factory test reports and certified listings on file.

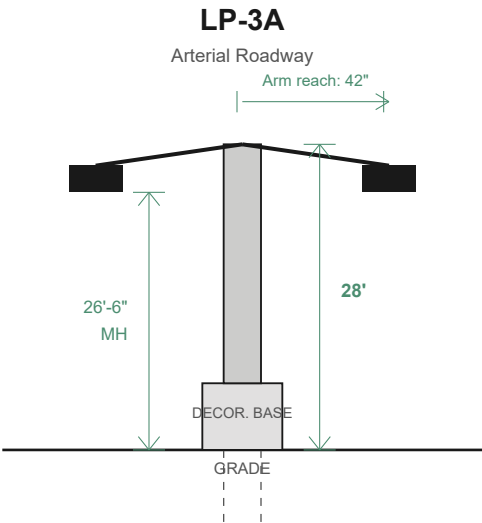
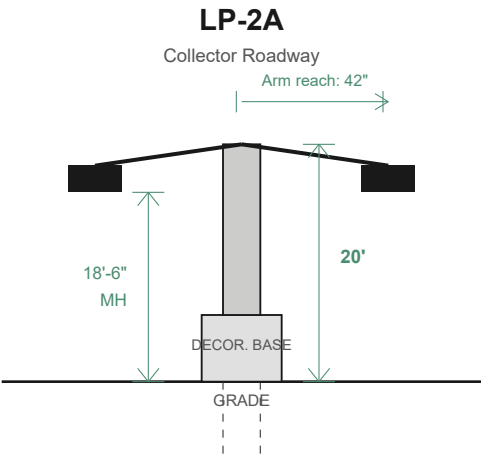
SYSTEM CONFIGURATIONS

LP-2A COLLECTOR / LP-3A ARTERIAL — SARATOGA SPRINGS STANDARD 05540

PROJECT: Municipal Road Lighting — Saratoga Springs, UT

TYPE: Street Light — LP-2A / LP-3A

DATE: March 2026



BILL OF MATERIALS — BOTH CONFIGURATIONS

Item	Description	Model / Catalog No.	LP-2A Qty	LP-3A Qty
1	Decorative LED Pendant Luminaire	Boris DSDP25 / 80 PLED / 3000K / 120V	2	2
2	Double Pendant Arm (180°)	Boris XAS-2-180 / Black RAL-9005-T	1	1
3	Round Straight Steel Pole — 20'	Boris RNTA205-188 / 5" / Q235 / Black	1	—
4	Round Straight Steel Pole — 28'	Boris RNTA285-188 / 5" / Q235 / Black	—	1
5	Decorative Cast Aluminum Base	Boris 2-pc wraparound / Black RAL-9005-T	1	1
6	LED Driver	Signify Xitanium XI180C090V285BSF1	2	2
7	Photocell Receptacle (NEMA 7-pin)	Included with luminaire	1	1

Specifications subject to change without notice. Consult factory for custom configurations. All fixtures must be installed per NEC and local codes.

COMPLIANCE MATRIX

SARATOGA SPRINGS CITY ENGINEERING STANDARDS 05530 (STREET LIGHTING) & 05540 (OUTDOOR LIGHTING)

PROJECT: Municipal Road Lighting — Saratoga Springs, UT

TYPE: Street Light — LP-2A / LP-3A

DATE: March 2026

The following matrix verifies compliance of the specified Boris Lighting products against Saratoga Springs Standard Technical Specifications January 2025, Sections 05530

LUMINAIRE — §05530 & §05540			
Type	Decorative LED pendant (§05530 2.1)	Boris DSDP25 — Solid State Decorative Pendant	COMPLY
Technology	LED solid state (§05530 1.7.A)	LED PLED® module array	COMPLY
Voltage	120V nominal (§05530 1.6.B / LP-1 Note 5)	120V / 60 Hz (rated 120–277V)	COMPLY
CCT	3000K (§05530 1.7.A — explicit)	3000K Warm White — specified for this project	COMPLY
CRI	Minimum 70 (§05540 2.1.A)	CRI ≥70 (high output PLED® module)	COMPLY
L70 Lifetime	≥50,000 hours	>90,000 hours (rated by manufacturer)	COMPLY
Power Factor	>0.90 (§05540 2.1.A.3)	PF 0.9945 (LM-79 tested at 120V)	COMPLY
Efficacy	Minimum 80 lm/W (§05540 2.1.A)	143.68 lm/W (LM-79 tested, Everfine GO-3000H)	COMPLY
Initial Lumens	Per photometric plan (§05540 2.1.A.3)	10,127 lm @ 70.49W / 120V (LM-79 tested)	COMPLY
Full Cutoff	IESNA full-cutoff required (§05540 1.8.A.7.c)	IESNA full-cutoff optics — zero uplight	COMPLY
Mounting Height	≤20 ft for full-cutoff fixtures (§05540 1.7.D.3.h.i)	20 ft (LP-2A) / 28 ft (LP-3A) — full-cutoff type	COMPLY
Surge Protection	Required (§05540 2.1.A)	20 kV external + 3 kV internal surge protector	COMPLY
Photocell	NEMA 7-pin receptacle (LP-1 Note 28)	7-pin photocell receptacle — photocell faces North	COMPLY
IP Rating	IP65 minimum (§05540 2.1.A)	IP66 per emitter (optical assembly)	COMPLY
Dimming	0–10V compatible (§05540 2.1.A)	0–10V dimmable driver (10%–100%)	COMPLY
Starting Temp	≤-20°F (-29°C)	-40°F (-40°C)	COMPLY
UL Listing	UL listed (§05530 1.5.B / §05540 1.5.B)	UL 8750 / CSA 250.13 — Signify Xitanium driver	COMPLY
NFPA 70 (NEC)	Conform to NFPA 70 (§05530 1.5.A)	NEC-compliant installation, bonded per §250.4	COMPLY
Finish	Black (LP-1 / LP-2A / LP-3A drawings)	TGIC polyester powder coat, Black RAL-9005-T	COMPLY
POLE — §05530 2.1 / LP-2A & LP-3A DRAWINGS			
Material	Steel pole (LP-2A / LP-3A drawings)	Q235 carbon steel, heat-treated (RNTA series)	COMPLY
Config LP-2A (Collector)	20 ft mounting height (LP-2A drawing)	RNTA205-188 — 20 ft nominal height	COMPLY
Config LP-3A (Arterial)	28 ft mounting height (LP-3A drawing)	RNTA285-188 — 28 ft nominal height	COMPLY
Wall Thickness	Min .188" (collector/arterial poles)	.188" nominal wall — RNTA series	COMPLY
Base Plate	Cast base with stiffeners (LP drawings)	Q235 cast steel, 6 vertical stiffeners	COMPLY
Bolts LP-2A	4 bolts 3/4"x18" at 90°, 12" circle (LP-2A)	Boris RNTA: 4 bolts 3/4"x18", 12" bolt circle	COMPLY
Bolts LP-3A	4 bolts 1"x36" at 90°, 12" circle (LP-3C)	Boris RNTA: 4 bolts 1"x36", 12" bolt circle	COMPLY
Hand Hole	Hand hole with cover (LP drawings)	18" opening with tamper-resistant grounded cover	COMPLY
Location	Centered in park strip (LP-1 Note 22)	Poles installed centered in park strip per plan	COMPLY
Finish	Black (LP-1 Note — Finish: Black)	TGIC polyester, Black RAL-9005-T	COMPLY
ARM — LP-2A & LP-3A STANDARD DRAWINGS			
Type	Pendant curved arm assembly (LP-2A / LP-3A)	Boris XAS single pendant arm (paired per config)	COMPLY
Arm Reach LP-2A	5 ft reach (LP-2A drawing)	XAS arm — 5 ft / 60" reach per city detail	COMPLY
Arm Reach LP-3A	7 ft reach (LP-3A drawing)	XAS arm — 7 ft / 84" reach per city detail	COMPLY
Finish	Matching luminaire (LP drawings)	Black RAL-9005-T — matches DSDP25 finish	COMPLY
ELECTRICAL INSTALLATION — §05530 3.2 / LP-1 NOTES			
Conduit	1.25" SCH40 PVC, 24" cover (LP-1 Note 12)	1.25" SCH40 PVC in sand bedding, 24" min cover	COMPLY
Wire	#6 AWG XHHW-2 copper (§05530 3.2.E / LP-1 Note 13)	#6 AWG XHHW-2 Cu — single conductor, no cable assy	COMPLY
Junction Box	Fused box ≤4 ft from pole base (LP-1 Note 18)	Bussman KTK-5 fuse, Carson Brooks clamshell box	COMPLY
Grounding	Bond all metal parts per NFPA 70.250 (LP-1 Note 19)	Ground stud in handhole — bonded per NEC §250	COMPLY
NEC / IBC	All work per NEC and IBC (§05530 3.2.I)	Full NEC / NFPA 70 compliant installation	COMPLY
Op. Demo	48-hour operational test (§05530 3.6.A)	Continuous operation verified with City Inspector	COMPLY
DECORATIVE BASE			
Material	Cast base — fluted decorative (LP drawings)	Min .225" wall, wraparound cast aluminum	COMPLY
Hand hole cover	Flush contoured cover (LP drawings)	Fluted cover, tamper-resistant hardware	COMPLY
Attachment	Set screws to shaft	(4) 1/4"-20 set screws to pole shaft	COMPLY

Sources: Saratoga Springs Standard Technical Specifications January 2025 §05530 & §05540; LP-1/LP-2A/LP-3A/LP-3C drawings; LM-79 test report (Everfine GO-3000H, Jan 26 2026); Signify Xitanium datasheet

Specifications subject to change without notice. Consult factory for custom configurations. All fixtures must be installed per NEC and local codes.

DSDP25

PHOTOMETRIC DATA — POLAR INTENSITY CURVES — IESNA LM-79 TEST REPORT — 3000K / 120V

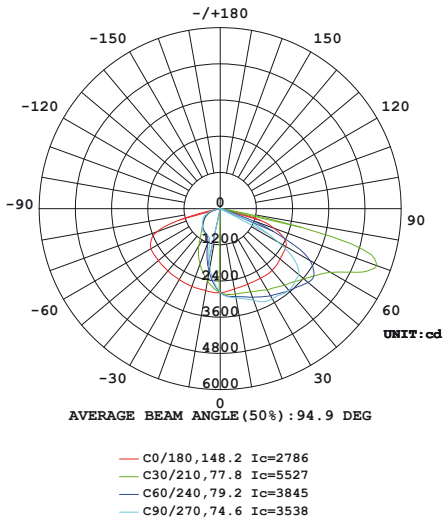
PROJECT: Municipal Road Lighting — Saratoga Springs, UT

TYPE: Street Light — LP-2A / LP-3A

DATE: March 2026

Polar intensity distributions below are derived from IESNA LM-79 photometric testing performed by Everfine (GO-3000H goniophotometer, Jan 26 2026) on the Boris DSDP25 luminaire at 120V / 3000K. The luminaire produces IESNA full-cutoff distribution — zero candela at or above the horizontal plane, confirming compliance with Saratoga Springs §05540 1.8.A.7.c.

LUMINOUS INTENSITY DISTRIBUTION DIAGRAM



LM-79 TEST SUMMARY — DSDP25 / 80 PLED / 120V / 3000K

TEST STANDARD	IESNA LM-79-08 (Approved Method for Electrical and Photometric Measurements of SSL Products)
TEST LABORATORY	Everfine (GO-3000H Goniophotometer) — January 26, 2026
LUMINAIRE MODEL	Boris DSDP25 — 80 PLED Module — 120V / 3000K (Warm White)
SYSTEM WATTAGE	70.49 W at 120.09V — measured at luminaire input
INITIAL LUMENS	10,127 lm (luminaire delivered lumen output)
EFFICACY	143.68 lm/W — system efficacy (luminaire)
POWER FACTOR	0.9945 (measured at 120V)
THD	<20% (driver harmonic distortion)
CCT (MEASURED)	3000K Warm White — per Saratoga Springs §05530 §1.7.A requirement
DISTRIBUTION TYPE	IESNA Full Cutoff — zero candela at or above 90° nadir
UPLIGHT (BUG)	B0 — no backlight, uplight, or glare above horizontal plane
COMPLIANCE	Saratoga Springs §05540 §1.8.A.7.c — All exterior lighting shall meet IESNA full-cutoff criteria

IES file (2026-1-26 _IESNA2002.IES) available for photometric software input. Full LM-79 signed test report available upon request.

Specifications subject to change without notice. Consult factory for custom configurations. All fixtures must be installed per NEC and local codes.

DSDP25

SOLID STATE DECORATIVE PENDANT LUMINAIRE — DSDP SERIES - PLED

PROJECT: Municipal Road Lighting — Saratoga Springs, UT

TYPE: Street Light — LP-2A / LP-3A

DATE: March 2026



DIMENSIONS

[Dimensional drawing — 20" W x 22" H]

PLED® MODULE OPTIONS

36 LED	48 LED	80 LED
40-122W	53-160W	88-232W
5,126-14,268 lm	6,835-19,024 lm	11,392-27,329 lm
@4000K / 120V	@4000K / 120V	@4000K / 120V

PRODUCT DESCRIPTION

The DSDP25 is a decorative LED pendant luminaire engineered for municipal street lighting applications. Upper housing is heavy gauge cast aluminum (min .125" wall, alloy >0.2% Cu) with integrated LED module seat. Lower housing is vented for convective cooling. Trulevel ball coupling mount is welded to the housing for quick leveling during installation. Designed for pendant arm mounting on collector and arterial roadways per Saratoga Springs standard details LP-2A and LP-3A.

FEATURES

- » Heavy gauge cast aluminum housing, corrosion resistant
- » PLED® optics — IP66 per emitter, H12 acrylic refractors
- » Field replaceable panels, rotatable in 90° increments
- » Tool-less access via Trulevel® quick-connect system
- » Constant current LED driver, PF >0.90, UL/cUL listed
- » 0-10V dimmable driver standard, 3KV internal surge
- » 20KV surge protector included for field installation
- » Photocell receptacle for dusk-to-dawn operation
- » 120V nominal per Saratoga Springs specification
- » Polyester powder coat — Black RAL-9005 standard

SPECIFICATIONS

HOUSING	Cast aluminum, min .125" wall (>0.2% Cu)
DIMENSIONS	20" W x 22" H (510 x 550mm)
E.P.A.	1.52 ft² (80 LED) / 1.10 ft² (60 LED)
LED MODULES	36 / 48 / 80 PLED® configurations
DRIVE CURRENT	350mA / 525mA / 700mA / 875mA / 1050mA
LED SOURCE	High output, 70 CRI minimum
CCT OPTIONS	3000K (WW) / 4000K (NW) / 5000K (CW)
VOLTAGE	120-277V, 50/60Hz (347-480V available)
DRIVER	Constant current, PF >0.90, 0-10V dimmable
L70 LIFETIME	>90,000 hours
STARTING TEMP	-40°F (-40°C)
SURGE PROT.	20KV external surge protector included
IP RATING	IP66 per emitter (optical assembly)
OPTICS	H12 acrylic, field replaceable, 90° rotatable
MOUNTING	Pendant arm — Trulevel® ball coupling
FINISH	TGIC polyester powder coat, 400°F bake
LISTINGS	UL / cUL recognized driver(s)

CERTIFICATIONS & COMPLIANCE

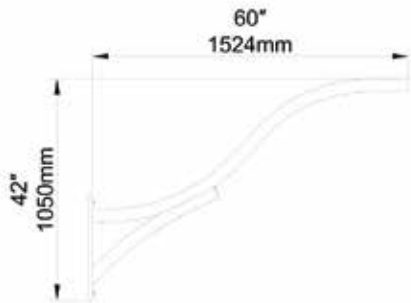
- » UL / cUL recognized LED driver(s)
- » NEC compliant installation per NFPA 70
- » Saratoga Springs City Standard 05540



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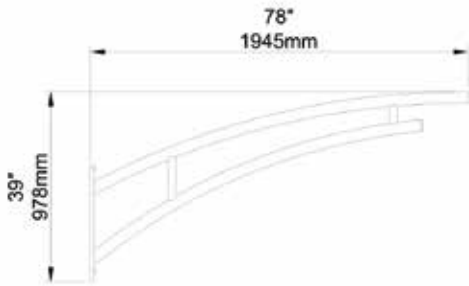
PENDANT MOUNTED ARM

PROJECT NAME:
Municipal Road Lighting in Utah



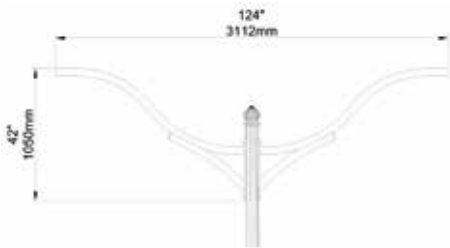
60"
1524mm

Height: 42" (1050mm)



78"
1945mm

Height: 39" (978mm)



124"
3112mm

Height: 42" (1050mm)



159"
3975mm

Height: 39" (978mm)

ARM SPECIFICATIONS

CONSTRUCTION	Cast aluminum, corrosion resistant
FINISH	TGIC polyester powder coat, 20 PSI wash at 140°F, 4-step media blast, 400°F bake
MOUNTING	Trulevel System® quick-connect assembly
ROTATION	±22.5 degrees
TORQUE	10 ft-lb
CONFIGURATIONS	XAS-WM / XAS-1 / XAS-2-180 / XAS-3-180 / XAS-3-120 / XAS-4-120 / XAS-4-90 / XAS-5-90
SELECTED CONFIG	XAS-1 (single arm) / XAS-2-180 (double arm per LP-2A/LP-3A)
SELECTED FINISH	Black RAL-9005-T

Specifications subject to change without notice. Consult factory for custom configurations. All fixtures must be installed per NEC and local codes.

○ ROUND STRAIGHT STEEL POLE

PROJECT NAME: Municipal Road Lighting in Utah
PROJECT TYPE: Street light
4" O.D. Top

FEATURES

Shaft

Shaft with an upper diameter of 4 inches and a lower diameter of 8 inches, fabricated from Q235 steel pipe. Heat-treated for enhanced strength and corrosion resistance. Features a covered hand hole with tamper-resistant bolt assembly. Base plate is cast steel construction.

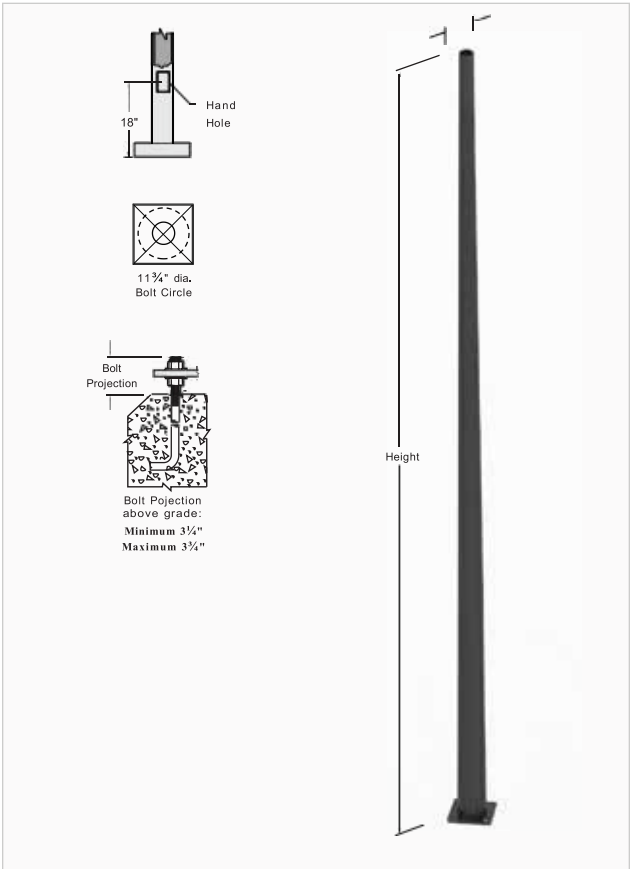
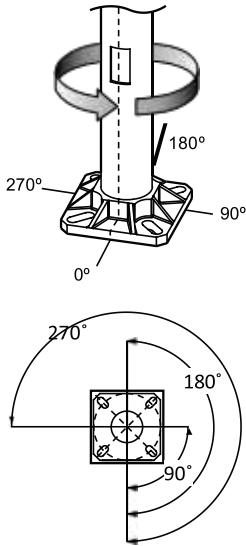
Base

Base is cast from Q-235 steel and heat-treated to achieve a corrosion-resistant state. The structurally designed base features six thick-wall vertical stiffeners. The base is retractable and is circumferentially welded to the shaft at the top, the outer side and the bottom of the base.

Light poles available in two styles with heights of 6 meters and 8 meters

RNTA5

Drilled Side Mount
Specify drilling location using codes below.



8" O.D. Base

BOLT CIRCLE

11-3/4" diameter
(4) 3/4" anchor bolts

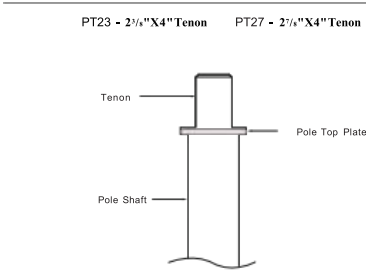
BOLT PROJECTION

Above grade:
Min 3-1/4" / Max 3-3/4"

TENON OPTIONS

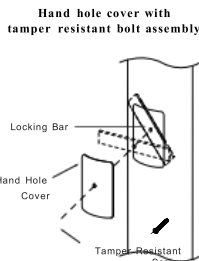
PT23 - 2-3/8" X 4" Tenon

PT27 - 2-7/8" X 4" Tenon



HAND HOLE COVER

Tamper resistant bolt assembly



Specifications subject to change without notice. Consult factory for custom configurations. All fixtures must be installed per NEC and local codes.

○ RNTA 5"

SPECIFICATIONS

Engineering Data

Maximum EPA - Square Feet

Model Number	Max. Fixture Weight	100 MPH	90 MPH	80 MPH	70 MPH
RNTA 105 - 125	175	8.7	10.8	13.9	18.8
RNTA 125 - 125	175	6.8	8.3	10.6	14.6
RNTA 145 - 125	175	4.4	5.5	7.1	9.9
RNTA 145 - 188	250	8.3	10.3	13.4	18.1
RNTA 165 - 125	100	3.2	4.0	5.2	7.3
RNTA 165 - 188	200	5.4	6.6	8.5	11.7
RNTA 185 - 125	100	2.2	2.9	3.8	5.6
RNTA 185 - 188	200	4.1	5.2	6.7	9.5
RNTA 205 - 188	200	3.1	4.0	5.2	7.6

All above design calculations are based on sustained
wind forces plus additional 1.3 wind gust
(Example: Pole rated at 80 MPH withstands 104 MPH gusts)

ORDERING INFORMATION

Spec/Order Example: RNTA205-188/1/RAL-9005-T/RBC

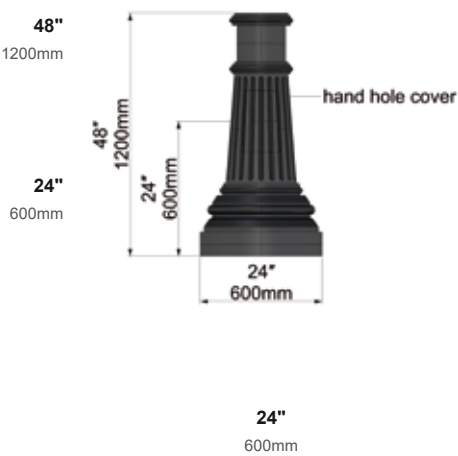
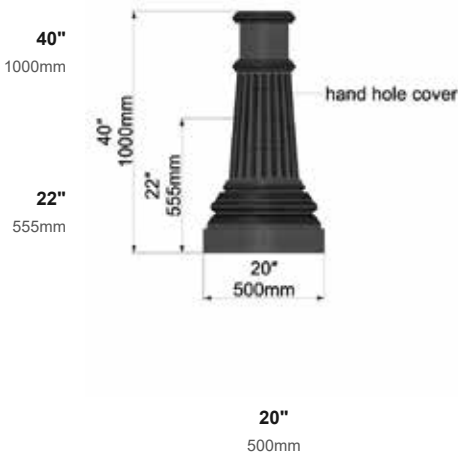
Pole Model Number - RNTA 5"				Mounting	Finish	Options
☐ Model	Ht	Wall	Bolt Circle	☐ PT23	☒ Black	☒ Round Base Cover
☐ RNTA 105-125	10'	.125	11-3/4"	2-3/8" X 4" Tenon	RAL-9005-T	RBC
☐ RNTA 125-125	12'	.125	11-3/4"	☒ PT27	☐ White	☐ 1/2" Coupling
☐ RNTA 145-125	14'	.125	11-3/4"	2-7/8" X 4" Tenon	RAL-9003-T	CPLN1/2
☐ RNTA 145-188	14'	.188	11-3/4"	☐ Other Tenon Mt	☐ Grey	☐ 3/4" Coupling
☐ RNTA 165-125	16'	.125	11-3/4"	Drill Mount	RAL-7004-T	CPLN3/4
☐ RNTA 165-188	16'	.188	11-3/4"	☒ 1	☐ Dark Bronze	☐ 2" Coupling
☐ RNTA 185-125	18'	.125	11-3/4"	☐ 2-180	RAL-8019-T	CPLN2
☐ RNTA 185-188	18'	.188	11-3/4"	☐ 2-90	☐ Green	
☒ RNTA 205-188	20'	.188	11-3/4"	☐ 3-90	RAL-6005-T	
				☐ 4-90		
				☐ 3-120		

Anchorage (all): 3/4" x 24" x 3" bolt pattern

Specifications subject to change without notice. Consult factory for custom configurations. All fixtures must be installed per NEC and local codes.

DECORATIVE BASE

Two-piece wraparound cast aluminum construction



SPECIFICATIONS

DECORATIVE BASE: TWO PIECE WRAPAROUND, CORROSION RESISTANT, DURABLE CAST ALUMINUM CONSTRUCTION. MINIMUM .225 WALL THICKNESS. BASE CONSISTS OF A SMOOTH TAPERED BOTTOM SECTION AND A DECORATIVE TAPERED FLUTED SECTION WITH EVENLY SPACED RAISED VERTICAL FLUTES. BASE IS FURNISHED WITH A CONTOURED, FLUTED FLUSH HAND HOLE AND COVER. HAND HOLE COVER SUPPLIED WITH TAMPER RESISTANT HARDWARE. (4) 1/4"-20 SET SCREWS PROVIDED TO SECURE DECORATIVE BASE TO SHAFT.

FINISH: POLYESTER POWDER COAT (SPECIFY COLOR.)

SCALE: 1/2" = 1'-0"

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DSDP25

LED & ELECTRICAL PERFORMANCE DATA — SYSTEM POWER AT 120V / 3000K

PROJECT: Municipal Road Lighting — Saratoga Springs, UT

TYPE: Street Light — LP-2A / LP-3A

DATE: March 2026

36 PLED MODULE

Drive Current	System Watts	Initial Lumens (4000K)
350 mA	40 / 46	5,126 - 5,882
525 mA	59 / 65	7,382 - 8,470
700 mA	79 / 85	9,362 - 10,742
875 mA	103 / 109	10,719 - 12,298
1050 mA	116 / 122	12,435 - 14,268

48 PLED MODULE

Drive Current	System Watts	Initial Lumens (4000K)
350 mA	53 / 59	6,835 - 7,843
525 mA	78 / 84	9,843 - 11,294
700 mA	105 / 111	12,483 - 14,323
875 mA	137 / 143	14,392 - 16,397
1050 mA	154 / 160	16,581 - 19,024

80 PLED MODULE

Drive Current	System Watts	Initial Lumens (4000K)
350 mA	88 / 94	11,392 - 13,071
525 mA	130 / 136	16,406 - 18,823
700 mA	174 / 180	20,805 - 23,871
875 mA	226 / 232	23,819 - 27,329

PHOTOMETRIC TEST SUMMARY (IESNA LM-79)

TEST DATE	January 26, 2026
TEST EQUIPMENT	Everfine GO-3000H V1
TEST VOLTAGE	120.21V / 0.5896A
SYSTEM POWER	70.487W
POWER FACTOR	0.9945
TOTAL FLUX	10,127 lm
EFFICACY	143.68 lm/W
BEAM ANGLE	94.9° (average 50%)
CIE CLASS	Direct (100% downlight)

LED DRIVER

MANUFACTURER	Signify (Advance)
MODEL	Xitanium XI180C090V285BSF1
OUTPUT POWER	180W maximum
INPUT VOLTAGE	120-277V AC, 50/60Hz
OUTPUT CURRENT	0.1A - 0.9A (SimpleSet)
POWER FACTOR	>0.90
DIMMING	0-10V, 10%-100%
SURGE	6kV combi-wave (CAT C)
TEMP RANGE	-40°C to +55°C
SAFETY	UL 8750, CSA 250.13

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